

Zarankiewicz for axis-parallel boxes

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- What about bipartite H ? ESS only tells $\text{ex}(n, H) = o(n^2)$!

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Theorem (Bukh, 2024)

$$z(n, n; s, t) = \Omega_t(n^{2-\frac{1}{t}}) \text{ if } s > 10^t$$

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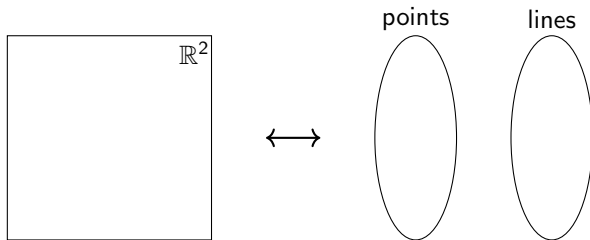
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Theorem (Chen–Liu–Ye, 2025+)

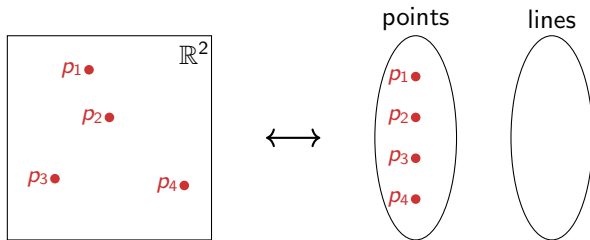
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Geometric incidences

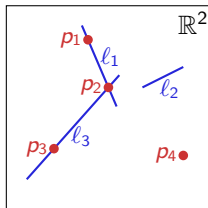
Geometric incidences



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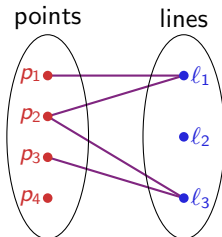
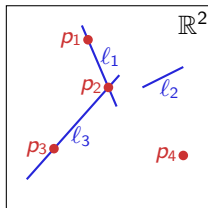
points



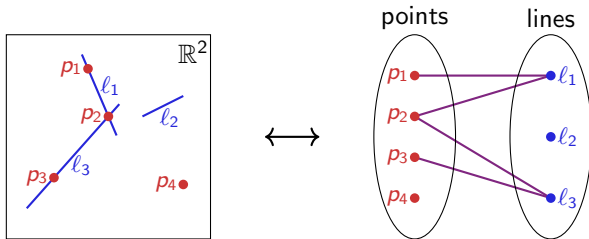
lines



Geometric incidences

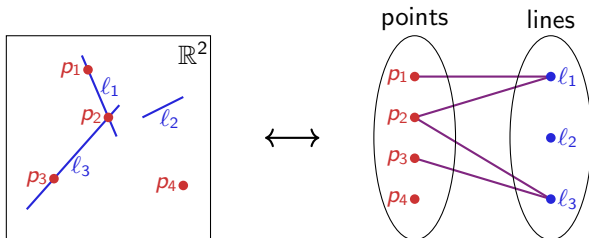


Geometric incidences



- point-line incidence graph G is $K_{2,2}$ -free $\xrightarrow{\text{KST}} e(G) = O(n^{3/2})$

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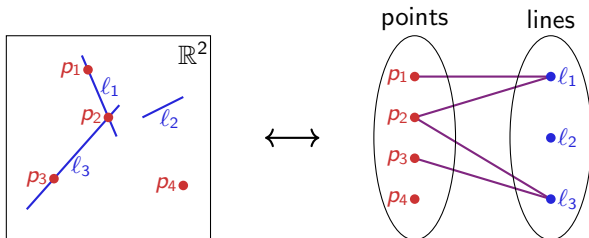


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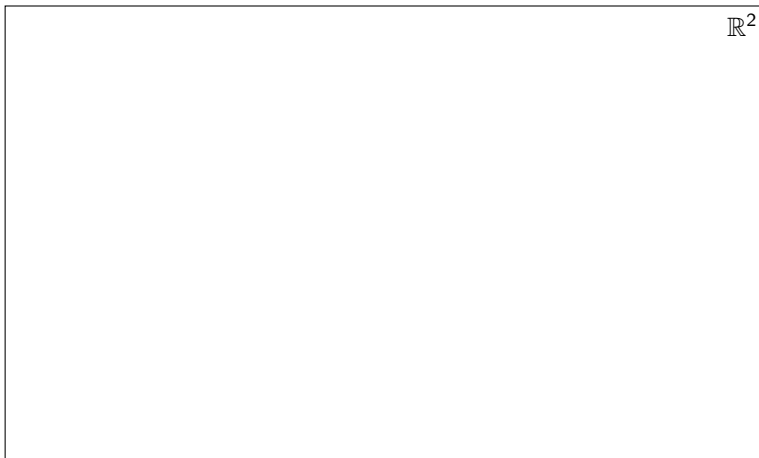
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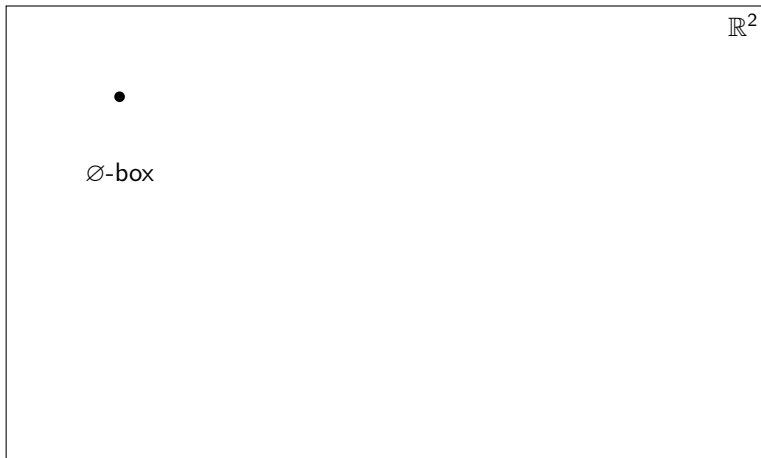
- This $O(n^{4/3})$ bound is tight!

Axis-parallel boxes

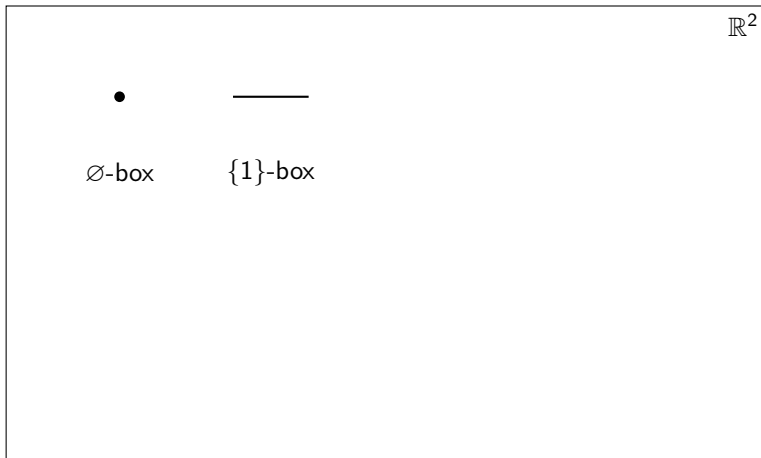
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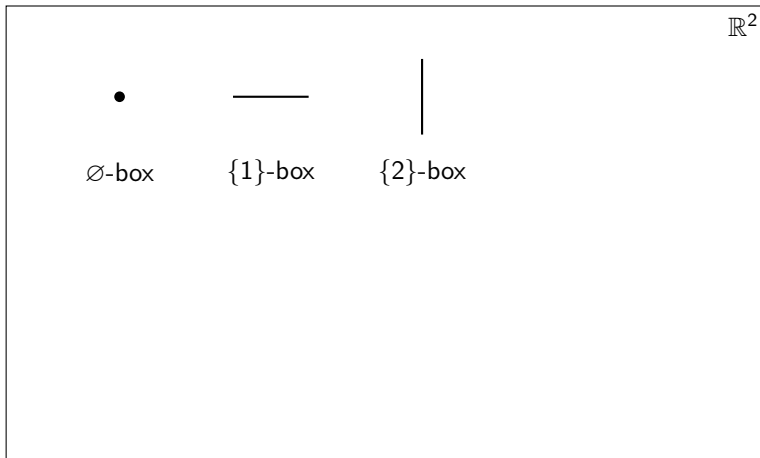
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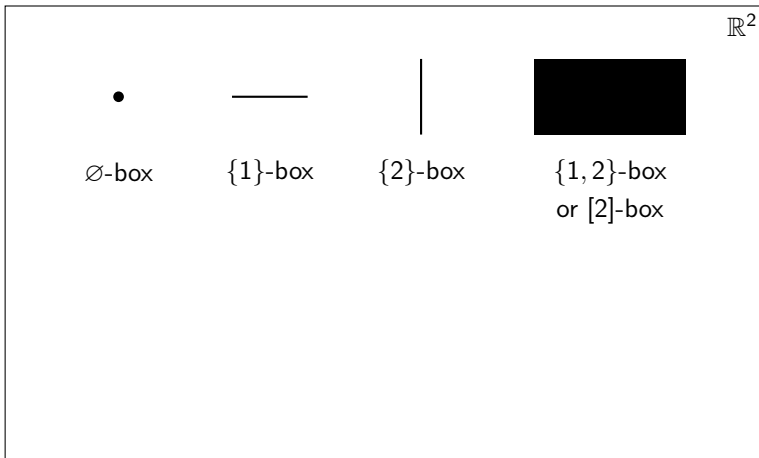
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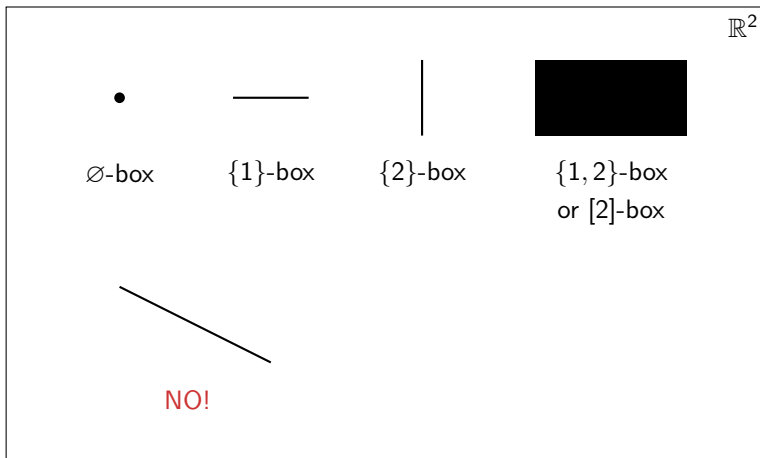
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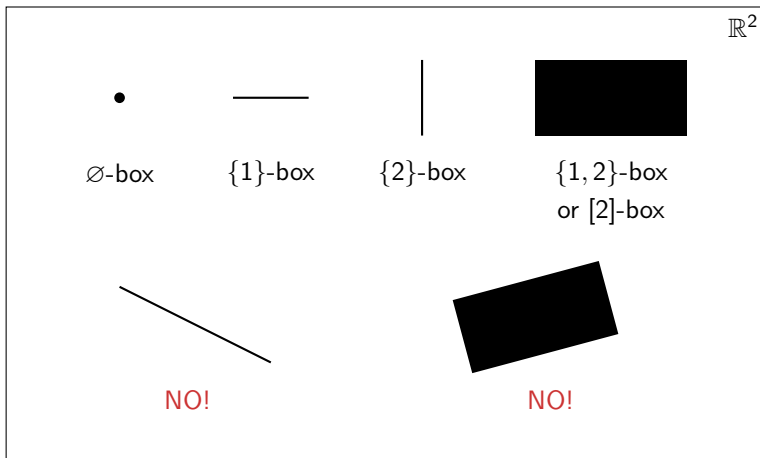
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Box Zarankiewicz: Assume that $\vec{F} = (F_1, \dots, F_r) \in (2^{[d]})^r$ is an r -direction-vector. Let $\mathcal{B}_i$ be an F_i -family of boxes in $\mathbb{R}^d$ with $|\mathcal{B}_i| = n$. Forbidding $K_{t, \dots, t}^{(r)}$, what is $\max \#$ edges in their incidence hypergraph $\mathcal{H}$?

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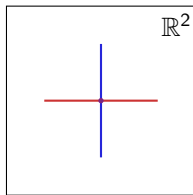
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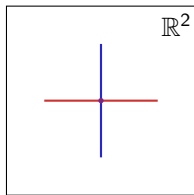
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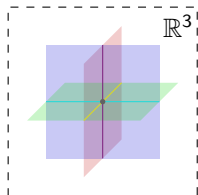
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- **Zarankiewicz** $z^r(n; t) \approx n^{r-\varepsilon}$ yet **Box Z.** $z_d^{(F_1, \dots, F_r)}(n; t) \approx n^{r-1}$

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- When does the extra logarithm factor happen?

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Theorem (Chao-D.-Liu-Shu-Wang, 2025+)

$$z_d^{\vec{F}}(n; t) = \begin{cases} \Theta_r(tn^{r-1}) & \text{if } \vec{F} \text{ is non-2-coherent,} \\ \Omega\left(tn^{r-1} \frac{\log n}{\log \log n}\right) & \text{if } \vec{F} \text{ is 2-coherent.} \end{cases}$$

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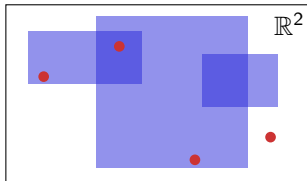
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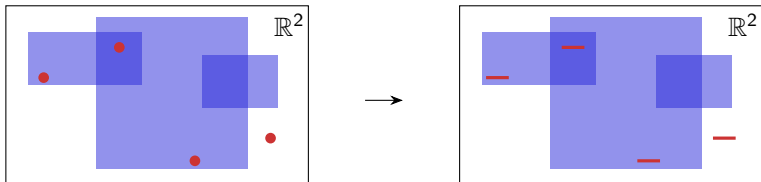


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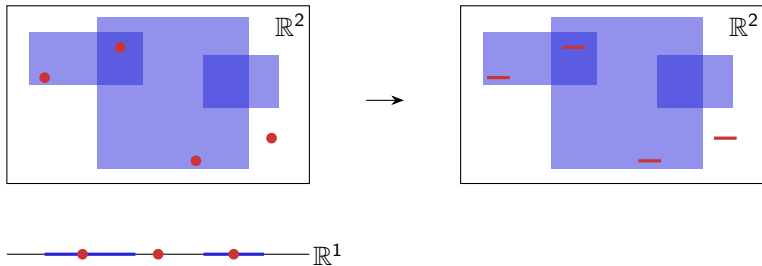


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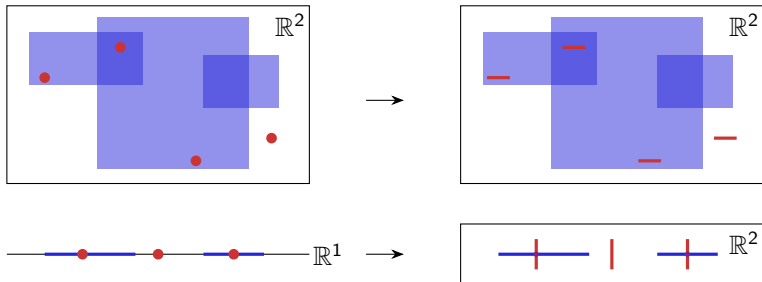


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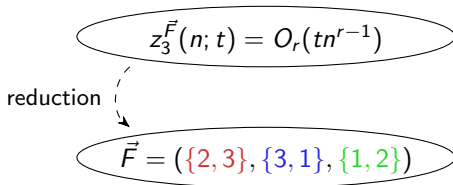


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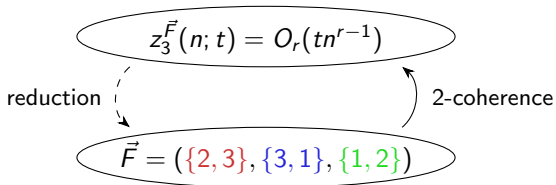
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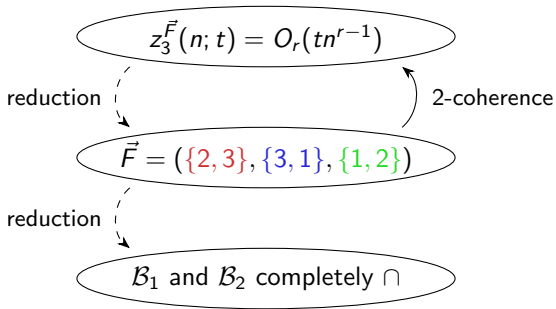
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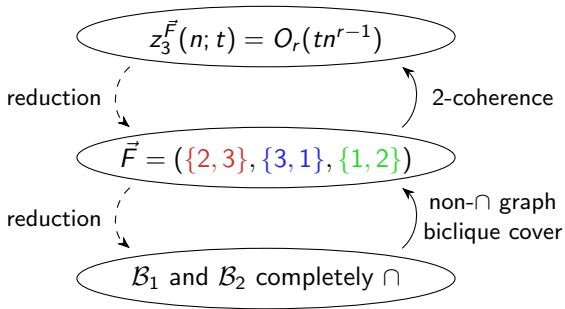
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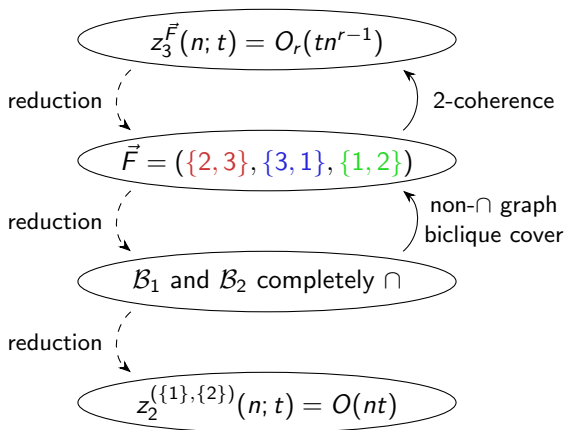
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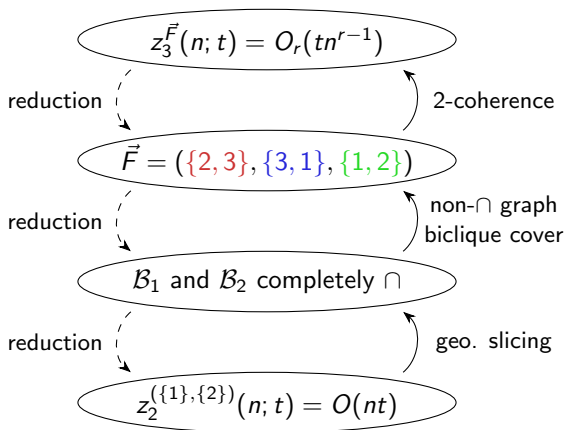
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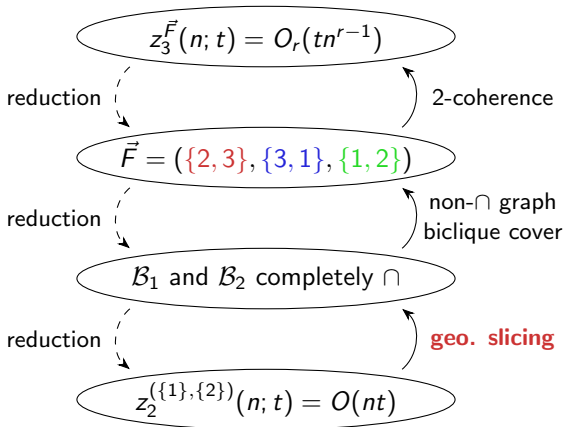
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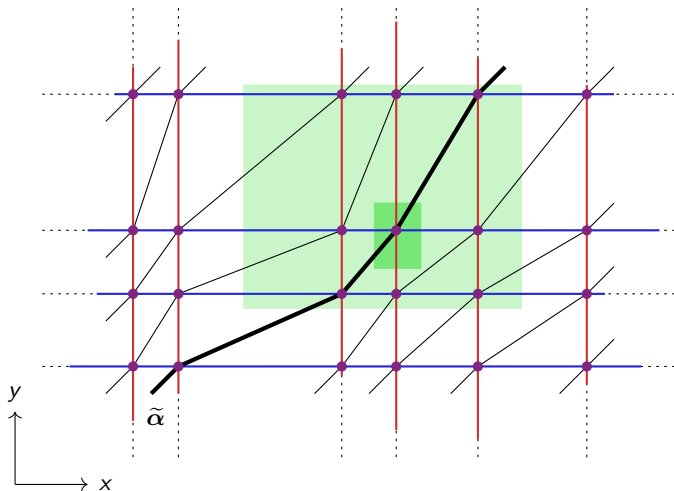
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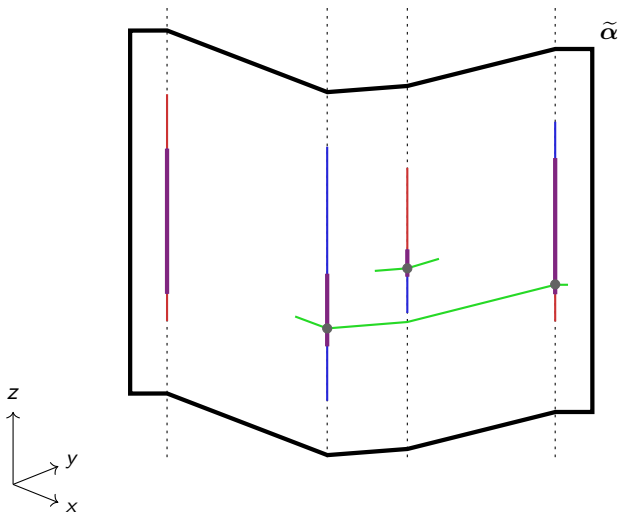
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Geometric slicing



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- Can you asymp. determine $z_4^{\{\{1,2\},\{3,4\}\}}(n; t)$? we know

$$\Omega\left(tn\left(\frac{\log n}{\log \log n}\right)^{\mathbf{1}}\right) \leq z_4^{\{\{1,2\},\{3,4\}\}}(n; t) \leq O\left(tn\left(\frac{\log n}{\log \log n}\right)^{\mathbf{3}}\right)$$

Open problems

Q: Can you find sharp asymptotics for $z_d^{\vec{F}}(n; t)$? *seems HARD*

- Can you asymp. determine $z_4^{\{\{1,2\},\{3,4\}\}}(n; t)$? *we know*

$$\Omega\left(tn\left(\frac{\log n}{\log \log n}\right)^{\mathbf{1}}\right) \leq z_4^{\{\{1,2\},\{3,4\}\}}(n; t) \leq O\left(tn\left(\frac{\log n}{\log \log n}\right)^{\mathbf{3}}\right)$$

